Robustness of Quantum Many-Body Scars

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Abstract

Quantum many-body scars (QMBS) are anomalous eigenstates that fail to thermalise in otherwise chaotic quantum systems, exhibiting long-time revivals instead of relaxing to thermal equilibrium as predicted by the Eigenstate Thermalisation Hypothesis (ETH). Since their experimental discovery in a 51-atom Rydberg simulator, a central open question has been how robust these states are once the system is perturbed away from fine tuned conditions. This project investigates the robustness of the Spin Helix State (SHS), an exact eigenstate of the integrable Heisenberg XXZ chain which is built from energy degenerate eigenstates in multiple magnetisation sectors. The SHS persists as a scar when the chain is made chaotic by adding longer ranged interactions with fine tuned anisotropy. The question of whether previously established scar robustness results apply to the SHS is raised due to its unusual structure. We investigate the SHS's stability under two different types of pertubation, the first being fine tuning of the anisotropy and the second generic symmetry-preserving potentials. This stability is quantified via fidelity susceptibility, bipartite entanglement entropy, and a kinetic energy observable. We find that the SHS consistently violates the ETH prediction of exponential scaling of fidelity susceptibility with system size. This confirms that indeed it behaves as a scar and not a generic state. The robustness of the scar depends heavily on the choice of system size and magnetisation sector, rather than just size like an ETH obeying state. Anisotropy perturbations destroy the scar smoothly following an approximate power law. In contrast, the generic potentials produce sharp spikes that lead to hybridisation at large perturbation strengths. These spikes are induced by near resonant states, and the SHS scar has been shown to partially recover after a near resonance rather than degrading monotonically. The results from this project imply that the SHS is a robust scar under certain choices of length, whose stability is dependent more on magnetisation sectors and less on size.

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1 Introduction

Chaos and integrability are fundamental topics in quantum information [13], that are frequently investigated today. When we move from classical to quantum dynamics, we no longer have the same diagnostics for what a chaotic system really is. This led to the discovery of RMT, and other related theories. The natural question that leads from discovering these dynamics is, what happens when one perturbs an integrable system, and how much is needed to make it chaotic? This has been explored greatly, and has been taken further to include discussion of how these dynamics change according to system size.

Deeply related to the concept of a chaotic system is thermalisation. The Eigenstate Thermalisation Hypothesis (ETH) is a statement about how quantum states in a chaotic Hamiltonian thermalise. Quantum scars are anomalous states, that disobey the ETH. In a spectrum that looks chaotic, where generic states are expected to thermalise over time, these scar states instead show revivals and defy our expectations. The question we pose in this project is, what happens when one perturbs a scar? In particular, we consider the SHS scar in the Heisenberg XXZ model.

In order to give context as to why quantum scars are so unexpected, we have to first establish a few different topics in quantum information. Firstly we'll discuss what we mean by chaotic and integrable, then we'll move to the ETH and finally we'll discuss what a quantum scar actually is and how to spot it. After this, we will discuss methods of perturbation and quantifying the robustness of states.

1.1 Integrable and Chaotic Quantum Systems

To begin this topic it's important to note a few classical definitions. Classically, a system is considered chaotic if it's phase space trajectory has an exponential sensitivity to small perturbations. Indicators like the Lyapunov exponent, which measures the rate of attraction to/ repulsion from a fixed point in state space [17], characterise the strength of this intensity and are often used to determine if a system is chaotic. Integrable systems on the otherhand are those where the equations of motion can be integrated to attain solutions in a closed form [8].

This becomes an issue when we start to try and define the concept of chaos in the quantum case. Firstly, due to the uncertainty principle we can't know a particle's position and momentum fully at a given time and trajectories are no longer straightforward to discuss. We can't rely on the Lyapunov exponent either as the Schrödinger equation is linear. In the end, we determine that we must need an entirely new theory to discuss quantum chaos. Wigner made the observation that statistical studies are a good place to start. He also noticed that when one chooses a generic basis for a chaotic Hamiltonian and focuses on a small energy window where the density of states is constant then it will look random [6]. Intuitively, this comes from the fact that the interactions in the chaotic system were complicated enough such that you end up with an effectively random matrix. This theory is called Random Matrix Theory (RMT) [9].

Although there is no actual definition for quantum chaos, the work of Wigner, Dyson and others allow us to use statistical studies based on RMT to examine if a quantum system is integrable or chaotic. In particular, the level spacings of integrable models are expected to show Poissonian statistics. This is in contrast to chaotic models whose level spacings exhibit level repulsion. The distributions for the chaotic case can be generated by assuming the entries in a Hamiltonian H are random variables (obeying the necessary

symmetries, i.e H is hermitian). Once the eigenvalues of H are calculated, an ensemble can be formed for their spacings, ω [1].

$$P(H) \propto \exp\left(-\frac{\beta}{2a^2}\mathrm{Tr}(H^2)\right) \tag{1.1}$$

Where in the above we have $\beta = 1$ in the GOE case and $\beta = 2$ for the GUE case. The GOE case refers to when time reversal symmetry is allowed and all entries are real, whereas the GUE case refers to when time reversal symmetry is not allowed and entries are complex.

To demonstrate the above point, we generate random matrices H_0, H_1, H_2 with the following rules.

$$H_0 = \mathrm{diag}(d_1, d_2, \ldots, d_m) \tag{1.2}$$

$$H_1 = \frac{1}{2}(M_R + M_R^T) \tag{1.3}$$

$$H_2 = \frac{1}{2}(M_C + M_C^\dagger) \tag{1.4}$$

Where in the above, $d_i \sim \mathcal{N}(0,1)\ \forall\ i \in 1, \ldots, m$, the same applies for all random variables in this instance. M_R is a random matrix with real entries and $M_C = M_{R1} + iM_{R2}$ is complex and built from two separate random real matrices. The first hamiltonian is integrable whereas the other two are expected to be chaotic. Through generating several samples of these matrices and performing a statistical study, we can observe that indeed we produce the expected distributions in fig.1.1.

To generate the aforementioned figure it's important to discuss how level spacing statistics studies are done. Firstly the eigenvalues are sorted $E_1 < E_2 < ... < E_n$, then the spacings are calculated as $\omega = E_{n+1} - E_n$. After this only the middle 50% values are taken, and we divide by the average to normalise.

$$s_n = \frac{E_{n+1} - E_n}{\langle E_{n+1} - E_n \rangle}. \tag{1.5}$$

where E_n are the ordered energy eigenvalues, such that $\langle s \rangle = 1$.

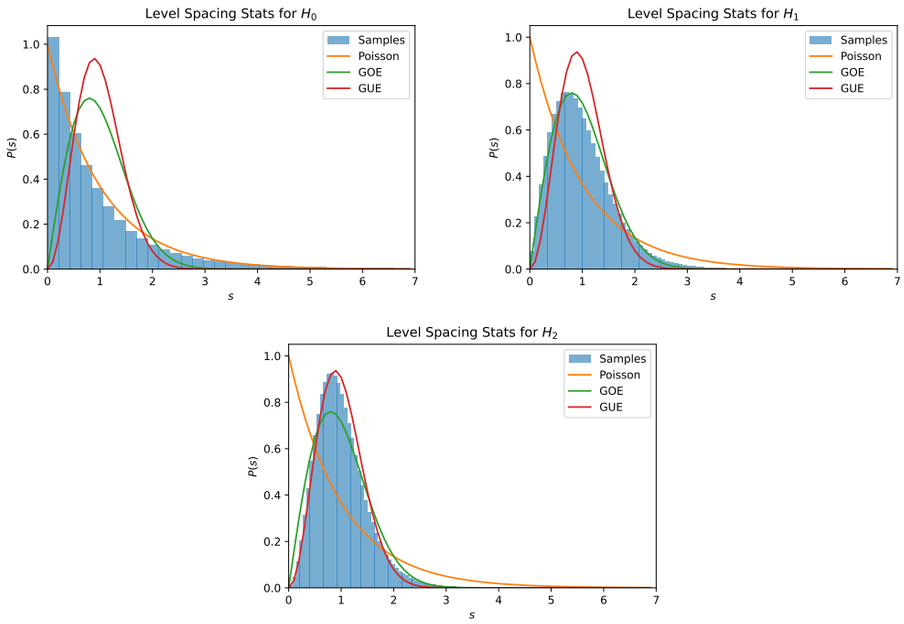


Figure 1.1: Level spacing statistics produced for a sample of 50,000 30x30 matrices. The integrable Hamiltonian clearly obeys Poissonian statistics whereas in the chaotic case we see GOE and GUE statistics respectively. Note that each plot has been given a uniform x axis.

1.2 Quantum Thermalisation and The ETH

It's helpful to begin discussion of this topic by defining a few things in statistical mechanics. When we have fixed energy, volume and particle number we can examine a system using the microcanonical ensemble. All valid states are weighted equally with probability $\frac{1}{\Omega(E,N,V)}$. Where $\Omega(E,N,V)$ is the total number of microstates. Then the thermal average of an observable A is given as

$$\bar{A}_{\mathrm{micro}} = \frac{1}{\Omega(E)}\sum_{i=1}^{N} A_i \delta(E_i, E) = \frac{1}{N}\sum_{i=1}^{N} A_i \tag{1.6}$$

If we have N, V, T fixed but variable E then we can utilise the canonical ensemble. In this case we define the ensemble average value of an observable A as,

$$\bar{A}_{\mathrm{canon}} = \frac{\mathrm{Tr}(Ae^{-\beta H})}{Z} \text{ where } Z = \mathrm{Tr}(e^{-\beta H}) \tag{1.7}$$

Now, it may be helpful to note that a system that might seem like it must be described by the canonical ensemble can instead be taken to the microcanonical one if we consider a window where the energy stays effectively constant.

In this paper, when referring to quantum thermalisation, I'll be using the definition from this review [5]. Consider an isolated system where the fluctuations in the systems energy grow smaller than the size of the system, and then prepare it in an initial state that's nonstationary. Then, any observable 'thermalises' if it relaxes to the microcanonical ensemble average over enough time and then fluctuates only a little around that value forever. In fewer words, this means that the long time average of our observable should equal the microcanonical prediction.

The concept of quantum thermalisation can then be linked directly to RMT. We will prove in the following that the RMT prediction for observables will result in one that also thermalises. This basically states that the diagonal elements $A_{\alpha\alpha}$ are independent of α and that $A_{\alpha\beta}, \alpha \neq \beta$ is exponentially small in the system size. These two statements can be condensed into a single equation,

$$A_{\alpha\beta} \approx \bar{A}\delta_{\alpha\beta} + \sqrt{\frac{\bar{A^2}}{N}} R_{\alpha\beta} \tag{1.8}$$

Where $R_{\alpha\beta}$ is a random variable with zero mean and a variance of 1. It is real for GOE statistics (where the variance of the diagonal components is 2) and complex for GUE statistics.

Now, assume we have a finite, nondegenerate, non-integrable Hamiltonian H, such that,

$$H|E_\alpha\rangle = E_\alpha|E_\alpha\rangle \tag{1.9}$$

We note that our solutions for $|E_\alpha\rangle$ must be separated by symmetry sector. Since the system is isolated, we can write the state of the system in terms of these eigenstates at any given time.

$$|\psi(0)\rangle = \sum_\alpha c_\alpha|E_\alpha\rangle \implies |\psi(t)\rangle = \sum_\alpha e^{-iE_\alpha t}c_\alpha|E_\alpha\rangle \tag{1.10}$$

Then taking the expectation value of any observable A at a given time,

$$\langle A(t)\rangle = \langle\psi(t)|A|\psi(t)\rangle = \sum_{\alpha,\ \beta} e^{-it(E_\beta - E_\alpha)}c_\alpha^* c_\beta A_{\alpha\beta}$$

Next we must calculate the long time average, $\bar{A}$. Before proceeding we're going to split up our sum into terms where $\alpha = \beta$ and those where $\alpha \neq \beta$

$$\langle\psi(t)|A|\psi(t)\rangle = \sum_\alpha |c_\alpha|^2 A_{\alpha\alpha} + \sum_{\alpha\neq\beta} e^{-it(E_\beta - E_\alpha)}c_\alpha^* c_\beta A_{\alpha\beta}$$

Notice that the first term has no time dependence, so we are going to focus our attention on the second term,

$$\lim_{t\to\infty}\frac{1}{T}\int_0^T \langle A(t)\rangle = \sum_\alpha |c_\alpha|^2 A_{\alpha\alpha} + \sum_{\alpha\neq\beta}\frac{ic_\alpha^* c_\beta A_{\alpha\beta}}{E_\beta - E_\alpha}\lim_{t\to\infty}\frac{e^{-it(E_\beta - E_\alpha)} - 1}{T}$$

We can show that the off diagonal terms go to zero by comparing the limit to a known converging one.

$$\lim_{t\to\infty}\left|\frac{e^{-it(E_\beta - E_\alpha)} - 1}{T}\right| \leq \lim_{t\to\infty}\frac{|e^{-it(E_\beta - E_\alpha)}| + 1}{T} = \lim_{t\to\infty}\frac{2}{T} = 0$$

Now, if the values $A_{\alpha\alpha}$are approximately constant within some energy window centered around the mean energy of the system, then we can write,

$$\sum_\alpha |c_\alpha|^2 A_{\alpha\alpha} \approx A\sum_\alpha |c_\alpha|^2 = A$$

Now note the definition of the microcanonical ensemble average,

$$\frac{1}{N}\sum A_{\alpha\alpha} \approx A\sum\frac{1}{N} = A$$

It's clear from the above that the microcanonical ensemble average is the same as the long time average of the expectation value. Which means that random matrix statistics implies thermalising observables.

The Eigenstate Thermalisation Hypothesis (ETH) is a statement concerning when isolated quantum systems exhibit thermal behaviour and can be described by thermodynamic ensembles [6]. Still following [5], we're going to describe the ETH in terms of an ansatz for the matrix elements of observables [16].

$$A_{\alpha\beta} \approx \bar{A}(\bar{E})\delta_{\alpha\beta} + e^{(-S(\bar{E})/2)}f_A(\bar{E},\omega)R_{\alpha\beta},\ \text{where}\ \bar{E} = \frac{E_\alpha + E_\beta}{2},\ \omega = E_\beta - E_\alpha \tag{1.11}$$

In the above we have that $S(E)$ is the thermodynamic extensive entropy, and all functions given an argument are smooth functions of said arguments. Once again $R_{\alpha\beta}$ is a random real or complex variable with zero mean and a variance of 1. It's worth noting that the ETH ansatz actually reduces to eq.(1.8) when considering a range of energies sufficiently small such that $f_A(\bar{E}, w)$ is constant.

We've thus far considered an ambiguous observable, but there are some constraints on the choice of observable for the ETH to hold. In particular, the ansatz eq.(1.11) is expected to hold for all observables for which statistical mechanics applies, and has been numerically verified for n-body observables where $n \ll N$.

1.3 Quantum Many-Body Scars

In 2018 an experiment was performed on a 51 qubit system simulating Rydberg atoms that discovered a surprising result [2]. While some initial states thermalized as expected, others showed long time revivals. This was due to a small number of anomalous non thermal eigenstates within the spectrum. This phenomenon was then later compared to chaotic stadium billiards [10]. In particular, it was compared to the phenomenon found when the billiard ball is replaced with a quantum particle. In the classical chaotic system, the ball launched at most angles will cover every possible trajectory in given time, bar a few angles that are an exception to this where the ball retraces the same path infinitely. In the quantum case it was found that the particles wave function rather than having equal probability everywhere, has a higher chance of being along the same paths as those in the retracing classical case. These were referred to as 'quantum scars', and as such the anomalous non thermalising states are now referred to as 'quantum many-body scars' (QMBS)[15].

To demonstrate some key facts about how these scars are spotted in practice, we look to an example Hamiltonian taken from [4],

$$H_{\text{toy}} = \Omega/2 \sum \sigma_i^x + \sum V_{i-1,i+2} P_{i,i+1} \tag{1.12}$$

$$V_{ij} = \sum_{\mu\nu} J_{ij}^{\mu\nu} \sigma_i^\mu \sigma_j^\nu \tag{1.13}$$

$$P_{i,i+1} = \frac{1 - \vec{\sigma}_i \cdot \vec{\sigma}_{i+1}}{4} \tag{1.14}$$

There are numerous ways to spot quantum scars in a system. Firstly, by definition the scars should not thermalise as expected, i.e they will show revivals. This means that the Fidelity $F = \langle\psi(t)|\psi(0)\rangle$ will return to 1 periodically throughout time, for a state $|\psi\rangle$ which is built primarily from scar states. Rather than picking a non eigenstate residing mostly in the scar basis, one could also demonstrate this by quenching the hamiltonian with some small potential. By this we mean sending $H' = H + \lambda V$, then $|E_{\text{quenched}}(t)\rangle = e^{-iH't}|E_{\text{scar}}\rangle$ where $|E_{\text{scar}}\rangle$ is a scar state of H. We will define two different states $|\psi_{\text{scar}}(0)\rangle, |\psi_{\text{gen}}(0)\rangle$, the first of which is deliberately chosen to lie mostly within the scar basis and the second chosen to be a generic thermal state. The resulting Fidelity calculations show exactly the expected revivals in fig.1.2b.

$$|\psi_{\text{scar}}(0)\rangle = |\uparrow\uparrow \ldots \uparrow\rangle = \bigotimes_{i=1}^{n} |\uparrow\rangle_i \tag{1.15}$$

$$|\psi_{\text{gen}}(0)\rangle = |\uparrow\downarrow\uparrow \ldots\rangle \tag{1.16}$$

Another method of spotting scars is recognising them as being anomalous low entropy states within the middle of the distribution [12]. By entropy we are referring specifically in this instance to the bipartite entanglement entropy [7]. This is computed by firstly cutting the chain into two subsystems A and B. When we trace over one of these systems we can pick up the reduced density matrix,

$$\rho_A = \text{Tr}_B[|\psi\rangle\langle\psi|] \tag{1.17}$$

The entanglement entropy is then defined as the Von-Neumann entropy of the reduced density matrix,

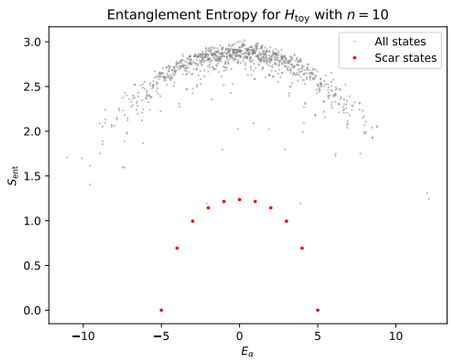
$$S = -\text{Tr}[\rho_A \ln \rho_A] = -\text{Tr}[\rho_B \ln \rho_B] \tag{1.18}$$

In practice, the entanglement entropy is often calculated using the singular value decomposition (SVD). The state $|\psi\rangle$ is first reshaped from a vector into an $N_A \times N_B$ matrix, where N_A and N_B are the Hilbert-space dimensions of subsystems A and B, respectively. Performing an SVD of this matrix gives us the Schmidt coefficients $\{\lambda_i\}$, from these the entanglement entropy can be calculated as

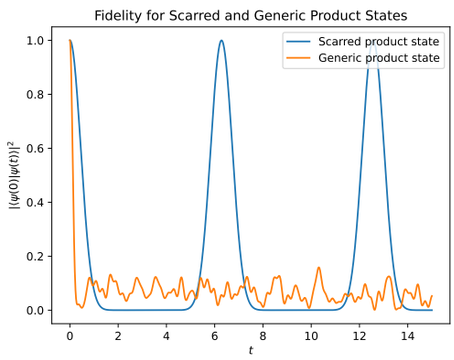
$$S = -\sum_i \lambda_i^2 ln(\lambda_i^2) \tag{1.19}$$

Where λ_i^2 are the eigenvalues of the reduced density matrix ρ_A.

In order for a state to be classified as a quantum scar it must have the traits we've described above, and it must lie within or close to the highest energy density region of the spectrum. When we analyse scars later we will be using the entanglement entropy again as a diagnostic for robustness.



(a)



(b)

Figure 1.2: The entanglement entropy of the eigenstates of H_{toy} plotted against the eigenvalues E_α clearly shows an evenly spaced distribution of scar states. These scar states are exactly those expected analytically.

1.4 Fidelity Susceptibility

Fidelity Susceptibility is a measure by which we can calculated how 'perturbed' a state is when the Hamiltonian is taken from $H \to H + \lambda V$. It is a quantity derived from first order time independent perturbation theory.

$$H' = H_0 + \lambda V \tag{1.20}$$

$$|n\rangle' = |n^0\rangle + \lambda|n^\lambda\rangle = |n^0\rangle + \lambda \boxed{\sum_{m\neq n} \frac{\langle m^0|V|n^0\rangle}{E_n^0 - E_m^0}} |m^0\rangle + O(\lambda^2) \tag{1.21}$$

Fidelity susceptibility χ_F is given by the boxed quantity. It is straightforward to show that the above can also be written in the following way, giving us two formulations for χ_F [20],

$$\chi_F[|n^0\rangle] = \lim_{\lambda\to 0} \frac{-2\ln|\langle n^\lambda|n^0\rangle|}{\lambda^2} = \sum_{m\neq n} \left|\frac{\langle m^0|V|n^0\rangle}{E_n^0 - E_m^0}\right|^2 \tag{1.22}$$

It's helpful at this point to introduce some short hand notation, we write $\langle m^0|V|n^0\rangle = V_{mn}$, $E_n^0 - E_m^0 = E_{nm}$.

The ETH also makes some predictions about how Fidelity should behave as a function of length. The ansatz in eq.(1.11) allows us to make the observation that $V_{mn} = e^{-S(\bar{E})/2} f(\bar{E}, \omega) R_{mn} \propto \exp(-S/2)$. We can couple this observation with eq.(1.8), which tells us via RMT that $E_{nm} \propto \exp(-S)$. Both of these approximations together with the fact that this S denotes the extensive entropy give us the approximation that $\chi \propto \exp(L)$ [18]. In theory then, we should expect that a generic thermalising state will exhibit this relationship whereas a scar state should not.

We can also define a 'stability to perturbations' from eq.(1.22). In particular, if a state is stable to perturbations then we expect $V_{mn} \approx 0$ as $E_{nm} \to 0$. The off diagonal matrix elements of V should suppress the large numbers generated by vanishing E_{nm}. This has been previously used to explore the stability of scars in the PXP model in [18], where it was shown that these scars exhibit exactly this suppression and $\chi_F \propto CL$, where C is some constant.

2 The Model

2.1 The Hamiltonian

There are multiple systems one can choose to look at and then embed scars into, however for our purposes we've decided to stick to one. In particular, we've chosen to look at the Heisenberg XXZ chain for variable interaction distance. Our primary reason for this choice was due to the plethora of literature behind this particular system. We note that we've also chosen periodic boundary conditions for our system.

$$H_k(\Delta) = \sum_{l=1}^{L} J(\sigma_l^x \sigma_{l+k}^x + \sigma_l^y \sigma_{l+k}^y + \Delta(\sigma_l^z \sigma_{l+k}^z - \mathbb{I})) \tag{2.1}$$

It's well known that this system is integrable for $k = 1$. We can break this integrability by simply adding higher H_k. Longer ranged interactions in the system induce chaotic behaviour.

$$H_1(J_1, \Delta_1) + \sum_{k>1} H_k(J_k, \Delta_k) \tag{2.2}$$

The XXZ hamiltonian has a total magnetisation symmetry, meaning $[H_k, H_0]$=0, and a spatial inversion (parity) symmetry $[H_k, P] = 0$. Since we've decided to narrow our focus to periodic boundary conditions we also pick up a translation symmetry, $[H_k, T] = 0$. The magnetisation symmetry is particularly interesting as

it allows the Hamiltonian to take a block diagonal form, each block labelled by a quantum number given by its total magnetisation. This fact becomes especially relevant later on when we discuss Spin Helix States.

The symmetries of H_k affect how statistical analysis must be done on the system. In order to calculate level spacing statistics and entropy/observable clouds correctly we must resolve these symmetries. This is done by restricting the eigenbasis to a specific symmetry sector. If this is not done, we will visualise multiple distributions. We have demonstrated the breaking of integrability in fig.2.1 by resolving these symmetries fully.

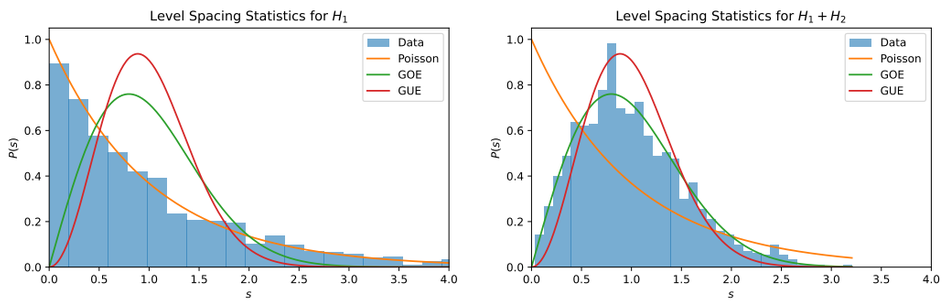


Figure 2.1: Level spacing statistics for H_1 and $H_1 + H_2$ demonstrating the breakdown of integrability as next to nearest neighbour interactions are added. In this figure the system has been set to $L = 19, J = 1, \gamma = \frac{2\pi}{L}$.

Calculations can indeed be made more complex due to these symmetries, but the computational advantage makes exploring the system much more feasible. Resolving the symmetries of the Hamiltonian reduces the Hilbert space size massively from the original 2^L, and many important calculations can be done within just one symmetry sector. This allows us to push further with L towards the thermodynamic limit. Another important point is that we now also have a restriction on our choice of perturbation V. V must obey the same symmetries as our Hamiltonian, if it does not then it will mix interactions between eigenstates of different sectors and analysis within an isolated sector is no longer possible.

2.2 Spin Helix State

The Spin Helix State (SHS) is an interesting example of an eigenstate of $\sum H_k$ under certain conditions. It's interesting primarily because it doesn't exist within one singular total magnetisation block, rather it's an eigenstate built from degenerate states across all magnetisation blocks [14]. It's of particular interest to us because it's a candidate for a scar state of the $H_1 + \sum H_k$ Hamiltonian. This is because although integrability is broken by the introduction of longer range interaction terms under fine tuning of Δ the SHS can persist unchanged.

The SHS state for the closed Hxxz chain is given by,

$$|SHS(\theta_0, \phi)\rangle = |\theta_0, \phi\rangle_1 |\theta_0, 2\phi\rangle_2 ... |\theta_0, (N-1)\phi\rangle_N \tag{2.3}$$

$$\tag{2.4}$$

Where in the above $|\theta_0, \phi_0\rangle = [1, e^{\theta_0 + i\phi}]^T$. The SHS is an eigenstate of any H_k as long as $\phi = \pm\gamma$ and $\gamma N = 2\pi m, m \in \mathbb{Z} \implies \gamma = \frac{2\pi m}{N}$. This γ is directly connected to H_k by definition, $\Delta = \cos(\gamma)$. The SHS remains an eigenstate for the chaotic system we created back in eq.(2.2) as long as $\Delta_k = \cos(k\gamma)$ [11].

The name 'Helix' can be intuited from its representation on the Bloch sphere. The vector at each lattice site (qubit) on the Bloch sphere is seen in fig.2.2 from [19]. As the length of the chain is traversed, one can notice the vectors trace out the shape of a helix.

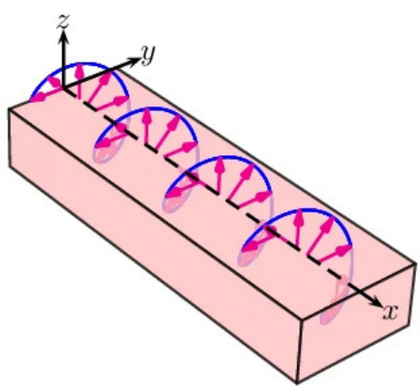


Figure 2.2: A diagram taken from [19] that elegantly shows the helix formed by the Bloch vectors at each lattice site.

At this point it's helpful to introduce a shift to our Hamiltonian such that the SHS's eigenvalue goes to 0. This can be helpful to reliably spot the degenerate eigenstates with which the SHS is composed of. We have this covered already in our H_k, as this is why we introduced H_k originally with a shift of $-J\Delta$ from the usual Heisenberg XXZ definition.

In order to break the SHS into a tower of scarred eigenstates,i.e break the energy degeneracy among its components in multiple magnetisation sectors, it can be helpful to add a total magnetisation term,

$$H_0 = \sum_{k=1}^{L} \sigma_k^z \tag{2.5}$$

This can also be added in order to quench the system and visualise the dynamics of the SHS.

2.3 Perturbations

The end goal of this project is to test how stable these scars are to perturbations. We define two seperate types of perturbation, the first being of fine tuning and the second of an added potential. In the case of fine tuning we perturb the anisotropy term Δ_k, as specific values of this are needed to keep the scar from being affected as higher H_k are added. This is of course very inconvenient in a practical setting, where such precision isn't possible. Although the perturbation of the anisotropy can be described as an added potential, we separate its discussion from the case where we include an unrelated Hamiltonian just to add noise. We have some restrictions on what these can be depending on what dynamics we choose to focus on.

We introduce two Hamiltonians, H_{par} and H_{DM}. The first of which has been chosen because it conveniently shares all symmetries with our Hamiltonian without commuting with it. The second has been chosen because there is already some literature on how it affects the SHS. Interestingly, strong Dzyaloshinskii-Moriya Interactions (DMI) have been shown to enhance the robustness of the SHS in one direction and weaken it in the other [21].

$$H_{\text{par}} = \sum_{i=1}^{L} \left(\sigma_i^+ \sigma_{i+1}^- \sigma_{i+2}^+ \sigma_{i+3}^- + \sigma_i^- \sigma_{i+1}^+ \sigma_{i+2}^- \sigma_{i+3}^+\right) \tag{2.6}$$

$$H_{\text{DM}} = \frac{iD}{2} \sum_i \left(\sigma_i^+ \sigma_{i+1}^- - \sigma_i^- \sigma_{i+1}^+\right) = D \sum_i \left(\sigma_i^x \sigma_{i+1}^y - \sigma_i^y \sigma_{i+1}^x\right) \tag{2.7}$$

We can also as mentioned previously describe the perturbation to the anisotropy in terms of an added potential V_Δ^k. This is useful to us later when we consider Fidelity Susceptibility.

$$V_\Delta^k = \sum_{l=1}^{L} \sigma_l^z \sigma_{l+k}^z \tag{2.8}$$

Our plan for testing the robustness of the SHS is shown schematically in fig.2.3. We aim to perturb the SHS scar in particular, and investigate the dependence of its robustness on L and the added H_k terms.

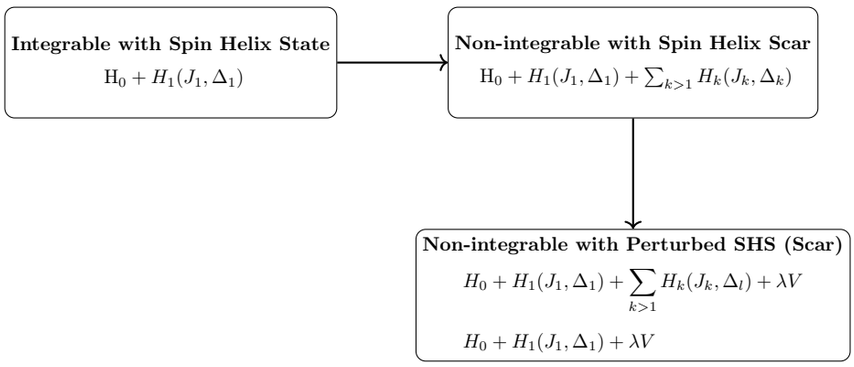


Figure 2.3: A diagram explaining our general strategy for perturbing the SHS. This includes the main topics we will cover in results.

3 Results

In general since we use quspin throughout the project we have some short hand for talking about symmetry sectors. We define N_{up} as the number of up spins, which determines the magnetisation sector. We also have kblock referring to the chosen momentum, and pblock referring to the parity.

In the following, we place a few constraints on the choice of L, m and N_{up} such that the SHS can be appropriately considered a scar. Firstly, we demand that the scar's eigenvalue sit near that of the bulk of the spectrum, this means our choice of m must be roughy half of L. Our next constraint is that we need enough states to confidently do a statistical analysis within a given magnetisation sector, so we need $N_{\text{up}} \approx L/2$ as well. However, we choose $N_{\text{up}} \neq L/2$ for our analysis to avoid the extra spin inversion symmetry it induces on our hamiltonian. Our final and weakest preference is that $\gcd(L, m) = 1$, we ask for this to avoid any

possible extra symmetries or degeneracies.

We use three main diagnostic methods to determine the robustness of an SHS scar to perturbations, fidelity susceptibility, the energy eigenbasis kinetic energy diagonal elements, and finally the bipartite entanglement entropy. The latter two are discussed under the heading of 'Observables'. In addition to perturbing our given Hamiltonian, we also look at the relationship of robustness to chain length, L under our previously stated constraints.

3.1 Fidelity Susceptibility

Fidelity susceptibility in general has proved to be a useful tool for determining the strength of scar states. The behaviour of the SHS isn't entirely what would be expected of most scars. The primary reason for this is the dependence of the SHS on L, m and N_{up}. The fidelity, and therefore stability, of the scar is wildly different depending on the choice of these parameters. The advantage gained from this is that one can choose a particularly favourable component of the SHS with which to do analysis with. In fact, the weight of a component of the SHS in a particular magnetisation sector can also be increased by choice of $z = \theta_0 + i\phi_0$. On the other hand, it can be argued that purposefully choosing these states, therefore reducing the SHS to one component in that sector, is ignoring important results about the behaviour of the entire SHS.

It's important to clarify the nuances in the nature of high χ_F. As we discussed in section 1.4, χ_F can blow up due to large V_{mn}, small E_{nm} or both. Thermal states have a very large χ_F since random matrix results from V_{mn} do not suppress small E_{nm}. The type of large χ_F we're seeing with the SHS is importantly not from RMT, and we can see this in fig.3.1. It is also likely not from any hidden degeneracies. One possible reason for the interesting shape of these distributions is near resonances. By this we mean two states $|n\rangle, |m\rangle$ for which Enm is small, but not 0. Later, at finite λ, we suspect that these states go through an avoided level crossing. Observables are introduced in section 3.2 that provide more context to this proposed explanation.

A known computational advantage of the total magnetisation symmetry of H_k is its block diagonal form. In general, statistical analysis (of the type in this paper) is restricted to one of these magnetisation sectors. It is possible however to calculate the total χ_F across all sectors without dealing with huge matrices by taking advantage of this symmetry. The SHS can be written by definition as a sum of energy degenerate states in different magnetisation blocks [14], which allows us to state,

$$\chi_F^{\text{Total}} = \sum_{N_{\text{up}}} \chi_F^{N_{\text{up}}}$$

We can quickly derive this,

$$|\psi_{\text{SHS}}\rangle = E_{\text{SHS}} \sum_{N_{\text{up}},\alpha} |E_{N_{\text{up}},\alpha}\rangle \; = E_{\text{SHS}} \left(\sum_{\alpha} |E_{0,\alpha}\rangle + \sum_{\alpha} |E_{1,\alpha}\rangle + \cdots + \sum_{\alpha} |E_{L,\alpha}\rangle \right) \tag{3.1}$$

$$\chi_F^{\text{Total}} = \sum_{m \neq \text{SHS}} \left| \frac{V_{m,SHS}}{E_{\text{SHS},m}} \right|^2 = \sum_{m \neq n} \left| \frac{\sum_{N_{\text{up}}} \langle m^0 | V | E_{N_{\text{up}},n} \rangle}{E_{\text{SHS}} - E_m^0} \right|^2 = \sum_{N_{\text{up}}} \chi_F^{N_{\text{up}}} \tag{3.2}$$

This allows us to get data on how the total SHS state should behave at different lengths, despite not having to actually work with huge matrices. Then, it can be helpful when picking a nice magnetisation sector to also consider the dynamics of the entire SHS.

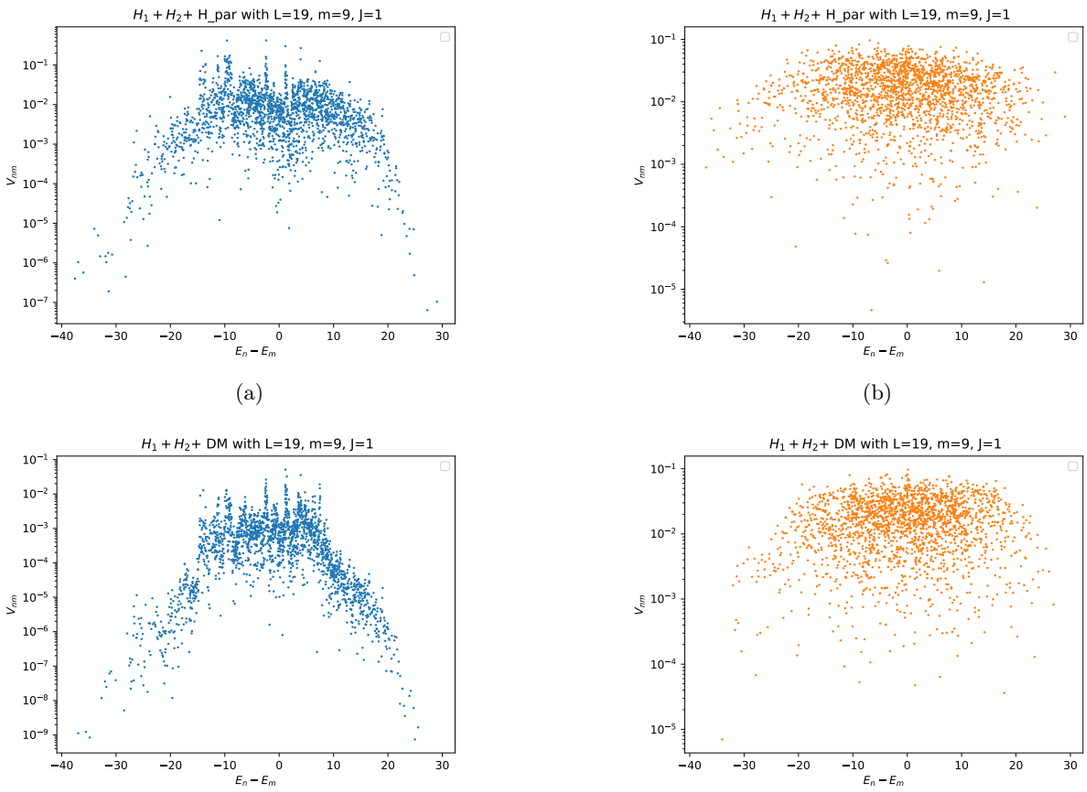


Figure 3.1: In (a) we have the matrix elements Vmn against Enm with $V = H_{\text{par}}$, $H = H_1 + H_2$, for a specific SHS scar. We compare this to (b) where the Vmn for a non-scar state obeys the ETH. The same comparison is being made for (c) and (d), however this time using $V = H_{\text{DM}}$, $H = H_1 + H_2$.

3.1.1 Anisotropy

Very few large Vmn/Enm results are needed to dominate the entire χ_F sum. We need only a few misbehaving states to send the fidelity susceptibility to even higher than thermal levels. Importantly, unless every symmetry is resolved it is common to come across extra degeneracies in the system. For example, in fig.3.2 we see that χ_F blows up for even lengths at $N_{\text{up}} = L/2$ due to the unresolved spin inversion symmetry. Due to time constraints, I did not resolve this symmetry and decided to focus on chains of odd L where possible, in which I did not find any degeneracies. It's also worth noting that likely due to the symmetric structure of the SHS there's a pattern where certain odd L tend to blow up. In particular if χ_F blows up at some N'_{up} then it also will at $L - N'_{\text{up}}$. It's also worth noting that these states are not within the same kblock, and in fact if the SHS in N'_{up} is at k' then $L - N'_{\text{up}}$ is at $L - k' mod(L)$. The L values with this property can change depending on the perturbation.

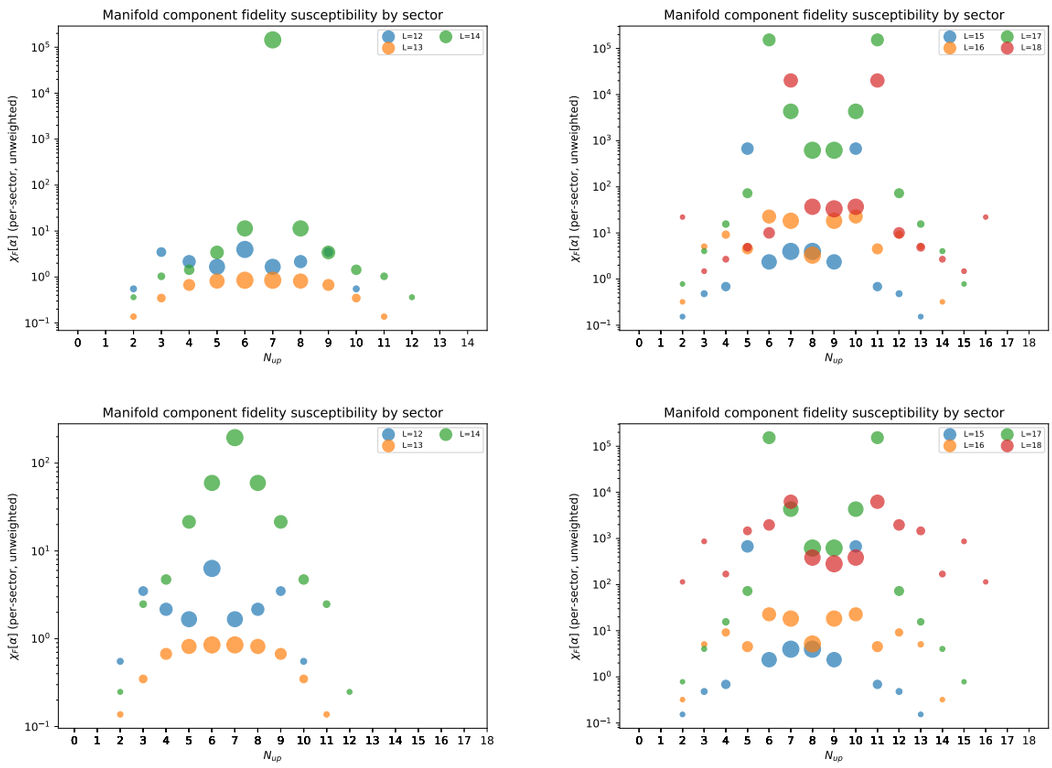


Figure 3.2: The SHS χ_F contribution in each N_{up} sector is shown with marker size $\propto$ the weight of the SHS in that sector. For (a), (b) we choose $m \approx L/2$ but don't demand that $\gcd(L, m) = 1$, whereas for (c),(d) we insist $\gcd(L, m) = 1$. In general the above values for L even and $N_{\text{up}} = L/2$ should be discarded as the large values are due to degeneracies from the spin inversion symmetry. These values have been kept in for completeness, but non degenerate perturbation theory is not valid there.

The values of χ_F^{Total} for $V = V_{\Delta,2}, V_{\Delta,2}$ are heavily dependent on the Hamiltonian being perturbed and the L, m pair chosen. Usually, one of the four plotted points will blow up above the ETH expectation. For $H_1 + H_2 + H_3 + \lambda V_{\Delta,2}$ 15 does, for $H_1 + H_2 + \lambda V_{\Delta,2}$ 17 does, and for $H_1 + H_2 + H_3 + \lambda V_{\Delta,2}$ 19 does as can be seen in fig.3.3. This aligns nicely with what we see in each χ_F^{Total} respective decomposition, where odd terms that blow up in a certain N_{up} sector usually come in pairs. This attribute of being in specifically arranged pairs makes this seem less random, and maybe related to something more fundamental about the system.

As we will later see, looking at both the χ_F^{Total} and its composition can be useful for studying the dynamics of these perturbations. They allow one to choose a convenient N_{up} sector while taking into account the weight of the SHS in that sector and its $\chi_F^{N_{\text{up}}}$ in comparison with the other sectors. This means we should be able to safely choose a nice system to work with later on when we shift our focus to observables.

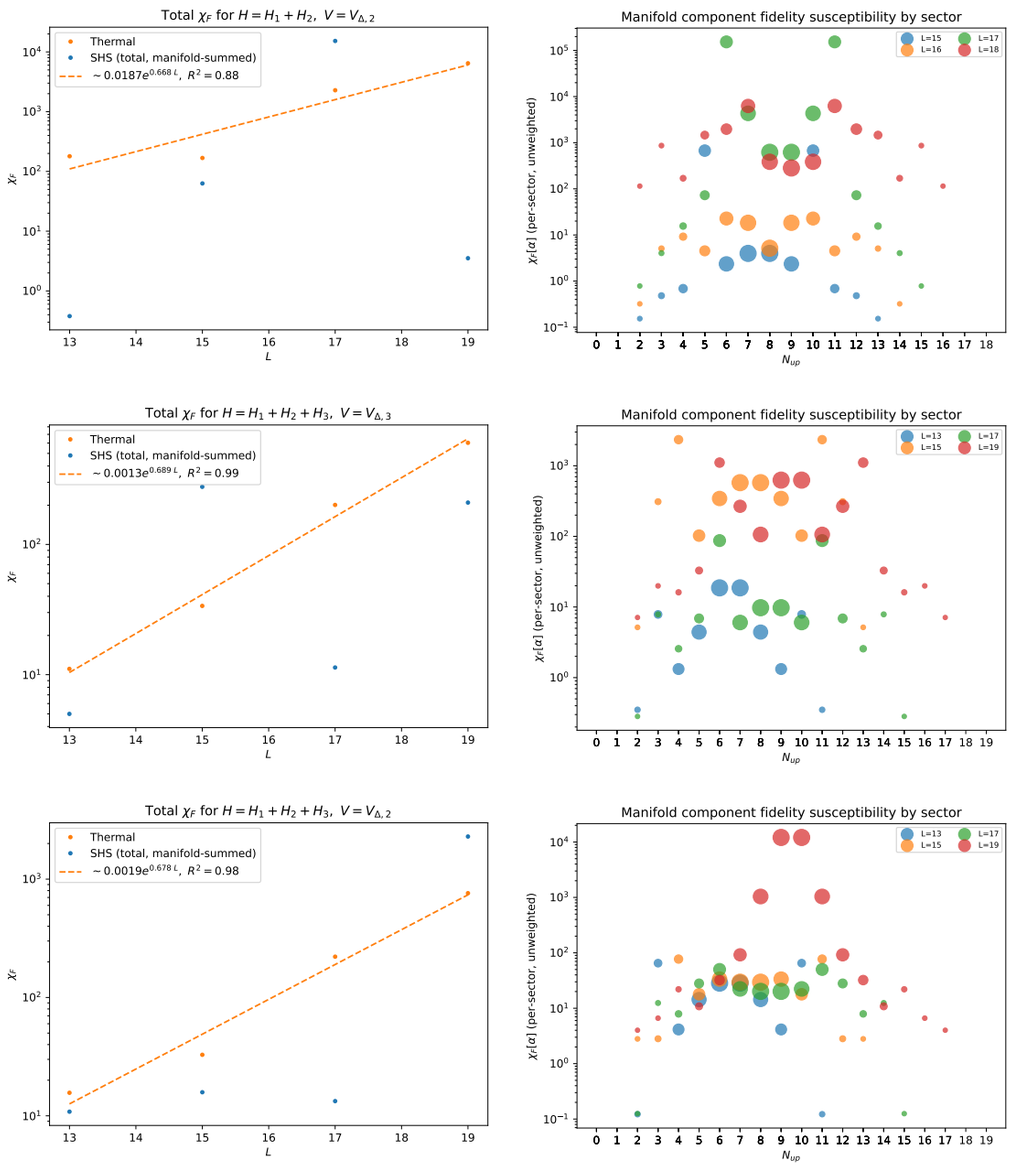


Figure 3.3: The total Fidelity Susceptibility is shown for odd L for $\gcd(L, m) = 1, m \approx L/2$ with several different configurations of H and V_Δ. The 'thermal state' values are computed as an average of non-SHS χ_F across a dense energy window. The ETH ansatz for the thermal case is verified by an exponential fit. Conversely, the SHS is shown to disobey the ETH prediction. The robustness of the SHS to perturbation varies wildly with the choice of L, m pair. On the right hand side of each total fidelity plot is its respective composition in terms of the N_{up} sectors.

3.1.2 Perturbations

For $V = H_{\text{par}}$ the resulting χ_F^{Total} in fig.3.4a matches nicely with what we saw previously, where the terms that blow up can be seen to be because of particular N_{up} sectors. It seems to act in a similar way to the $V = V_{\Delta,2}$ perturbation, with different strengths but the same basic profile. Once again it looks like the $L = 13, m = 7$ state remains the strongest, which is partially due to its smaller length, but may have some other mechanism keeping it robust as well. Moving to $V = H_{\text{par}}$, this is the first perturbation we've seen in this report where each χ_F^{Total} is below the expected ETH value. This could be due to selectively small V_{mn}, that cancel out contributions from the small E_{nm}.

Due to computational and time limitations it was not possible to calculate χ_F^{Total} for $L > 19$. However, this was achievable when restricted to a single N_{up} sector for each L. In order to do some further analysis, we look at $\chi_F^{N_{\text{up}}}$ with $N_{\text{up}} = L//2 - 1$, and we calculate this up to $L = 20$, shown in fig.3.5. The resulting plots mirror those in fig.3.4 quite nicely. This may be evidence to support that what one can learn from one total magnetisation sector can in cases be used to make predictions about the total SHS dynamics. When scaling up to 20 it's now also clear that the SHS with $L = 19, m = 8, 7$ are also relatively robust.

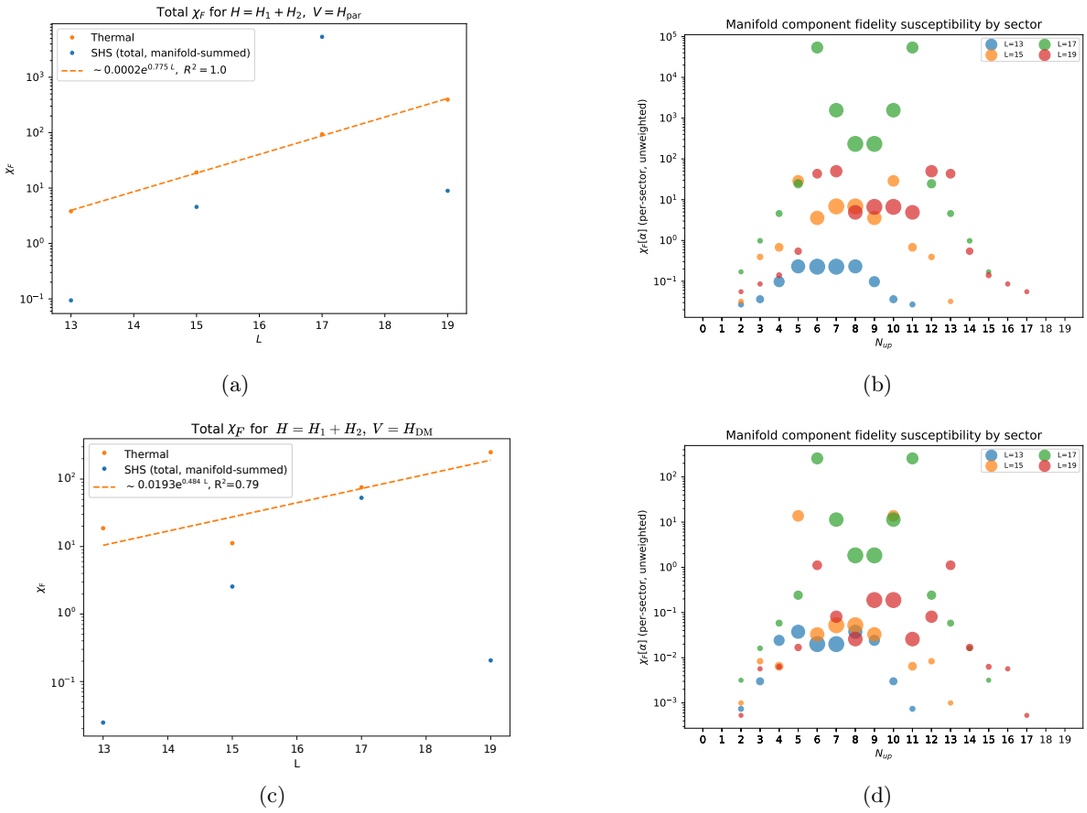


Figure 3.4: The fidelity susceptibility in (a) once again shows the ETH correctly predicts the thermal case but does not predict the SHS case. We can compare (a) with (b) to understand that the high χ_F, for example at $L = 17$ is due to contributions from specific N_{up}. In (c) we see the $V = H_{\text{DM}}$ perturbation is quite well behaved with lower χ_F than the ETH prediction in general.

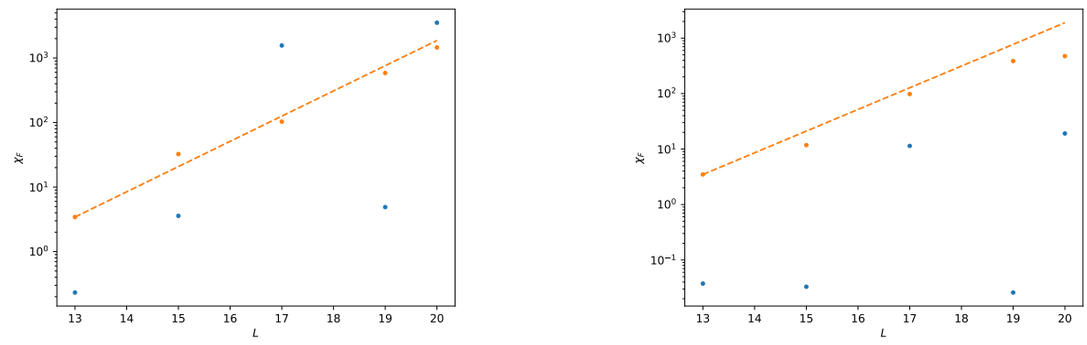


Figure 3.5: Looking at the Fidelity Susceptibility in N_{up} sector $= L//2 - 1$, in (a) for H_{par} and in (b) for H_{DM}. We've restricted to one N_{up} sector here to demonstrate the relationship between fidelity values and sector choice, as well as to push to $L = 20$ which was not computationally feasible for χ_f^{Total}.

3.2 Observables

There's two main observables we'll be using to analyse the SHS. The first is the bipartite entanglement entropy. As previously discussed, this is a trusted method in the literature to pick out a scar state as an anomalous low entropy state near the highest energy density. The second observable we'll look at is kinetic energy. We have not discussed it yet in this report, but it has been previously used in papers such as [3] to be a measure of integrability. As it turns out, this kinetic energy observable also shows the SHS as an anomalous state near the highest energy density, and it obeys all of the symmetries we seek to resolve. The comparison of these two observables and their dependence on λ will hopefully show some form of consistent pattern with the SHS.

$$T = \frac{1}{N} \sum_{n=1}^{N-1} (\sigma_n^x \sigma_{n+1}^x + \sigma_n^y \sigma_{n+1}^y) \tag{3.3}$$

The choice of our observables gives us a computational advantage again for higher length chains. The bipartite entropy can actually be considered in a single N_{up} sector, though it can't be calculated in the further kblock and pblock reduced form. It can be a slightly expensive calculation due to the SVD method

of calculating it and losing the advantage from translation and parity. The kinetic energy is a much cheaper calculation, it can be considered in a single N_{up} sector, kblock and pblock. Coupled with this, it doesn't use any particularly numerically intensive algorithm, it can be calculated through simple matrix multiplication. This allows us to push further towards the thermodynamic limit.

In general it appears as though the main thermal distribution appears more clumped as higher H_k terms are added, fig.3.6. In this section we also seek to probe this and consider if the SHS becomes weaker. It's not immediately clear, since yes the higher interacting terms should spread the perturbation along the chain faster, but there are also less outliers amongst the thermal states.

In this section we will measure perturbation of an observable by the quantity,

$$\delta T = \text{av}(\langle E_\alpha|T|E_\alpha\rangle) - \langle E_{\text{scar}}|T|E_{\text{scar}}\rangle \tag{3.4}$$

$$\delta S = \text{av}(S_\alpha) - S_{\text{scar}} \tag{3.5}$$

Where av denotes the average over a small window of high energy density states. When $\delta T, \delta S$ are both minimised we expect the scar to be destroyed. In general, we plot $\delta T(0) - \delta T(\lambda)$, since it makes observing this relationship more convenient. Tracking these scars as λ is increased is a separate challenge that will be discussed further in appendix A.

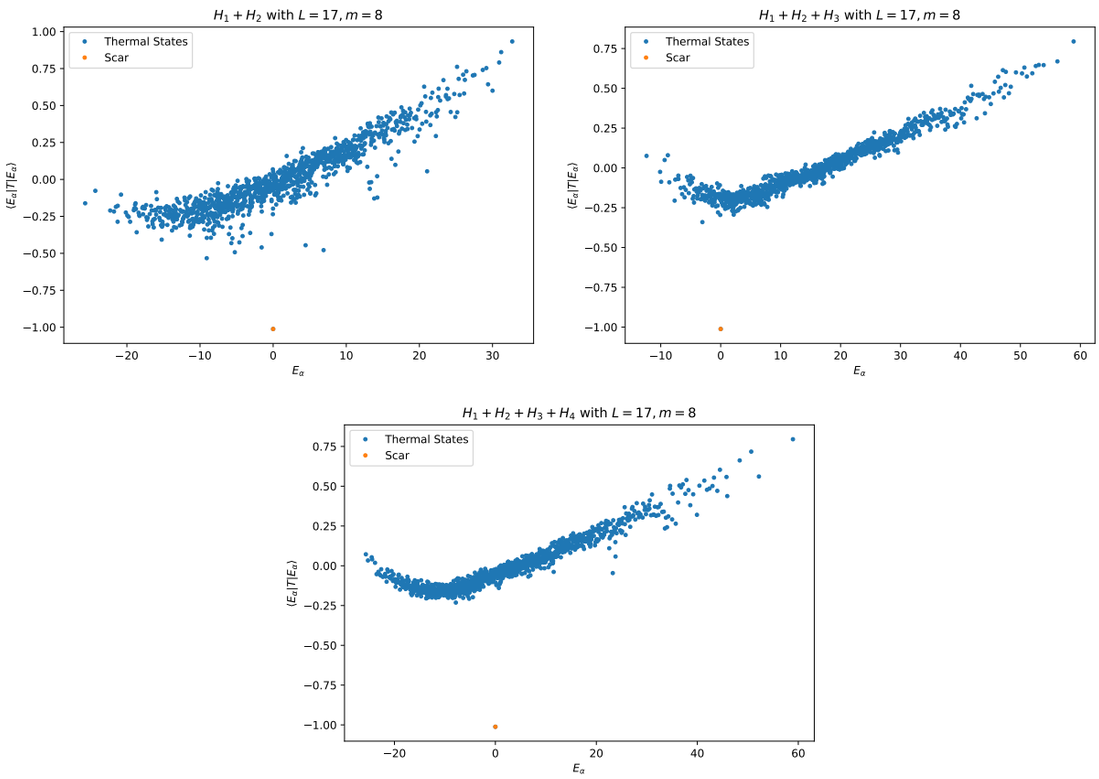


Figure 3.6: As we increase the the number of H_k we add to the original integrable H_1 it's clear that the thermal distribution becomes more distinct from the scar. This distinction makes the scar more anomalous below the thermal bulk.

3.2.1 Anisotropy

We perturb the Anisotropy with the same method used in the previous section, and we compare our predictions from χ_F^{Total} to how the state actually behaves. For this purpose, we choose a proven low fidelity susceptibility system $L = 13, m = 7, N_{\text{up}} = 7$. We make this choice not only because of the lack of variance of $\chi_F^{N_{\text{up}}}$ between different N_{up} sectors but also because we expect it to be easier to track.

Perturbing the Anisotropy such that $H \to H + \lambda V$ with $H = H_1 + H_2$ and $V = V_{\Delta,2}$ produces some relatively smooth results for $L = 13, m = 7, N_{\text{up}} = 7$ as seen in fig.3.7. This is true for both analysis on kinetic energy and on entropy. Looking at entropy we have the relationship, $\delta S(\lambda) \approx -0.305\lambda^{1.37} + \delta S(0)$, with $R^2 = 0.99$. This is quite a high R^2 value so we expect this to be relatively accurate, however, the smooth variation around the line fit could hint at a more complex relationship . In the kinetic energy case we have $\delta T(\lambda) \approx -0.563\lambda^{1.22} + \delta T(0)$, with $R^2 = 0.97$. The lower R^2 possibly comes from close resonance which may create the discontinuous jumps in the $\delta T(0) - \delta T(\lambda)$. Both $\delta T(\lambda), \delta S(\lambda)$ grow with a very similar power law, which is close to linear.

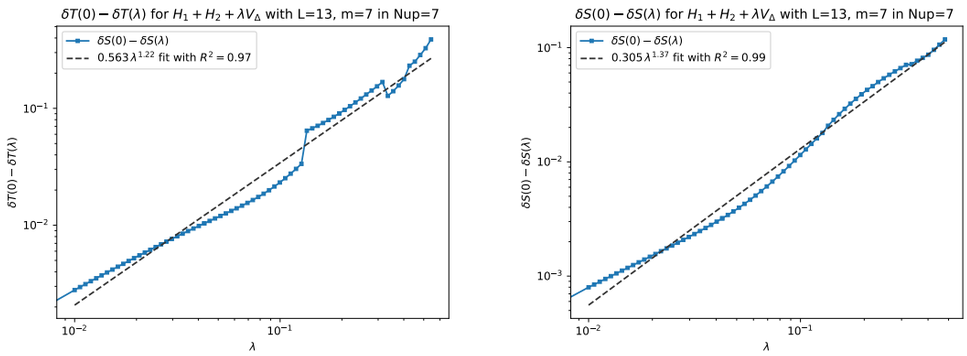


Figure 3.7: The perturbation $V_{\Delta,2}$ is considered for the $H_1 + H_2$ system. The relationship with lambda is very similar in both plots. Although the entropy appears to grow faster, it's a much smoother plot with a more accurate fit than kinetic energy so this might not mean the observables grow at inherently different rates.

When we switch to the Hamiltonian $H = H_1 + H_2 + H_3$, we now have two Δ_k to perturb. The $V = V_{\Delta,2}$ relationships remain very similar to the previously discussed case. We find that $\delta S(\lambda) \approx -2.45\lambda^{1.47} + \delta S(0)$ and $\delta T(\lambda) \approx -6.4\lambda^{1.69} + \delta T(0)$, with the R^2 values remaining roughly the same. Clearly the addition of a further chaos inducing term has pushed the SHS scar to hybridse faster. This is consistent with the intuition that longer range interactions can provide additional ways for the perturbation to couple distant regions of the system. This could possibly allow a perturbation's effect to propagate more rapidly.

The $V = V_{\Delta,3}$ perturbation is qualitatively different to the $V = V_{\Delta,2}$ case as shown in fig.3.8. The characteristic spike could possibly be from an avoided level crossing. When we perturb the system with other potentials in section 3.2.2 a similar result is investigated further in a system with worse behaviour. The scar as measured by $\delta T(\lambda)$ and $\delta S(\lambda)$ seems to hybridise at $\sim 4 \cdot 10^{-1}$. This is longer lasting than the $V = V_{\Delta,2}$ perturbation on the same Hamiltonian. Despite being longer lasting, it still survives for less λ than the $H = H_1 + H_2,\ V = V_{\Delta,2}$ case.

Perturbing the Anisotropy has surprisingly smooth results overall. There doesn't seem to be a huge amount of issue with resonance, and in the $V_{\Delta,2}$ case for both $H = H_1 + H_2$ and $H = H_1 + H_2 + H_3$ we can derive a power law relationship. In general the kinetic energy and the bipartite entropy tend to roughly agree on the relationship with λ. The kinetic energy however has worse R^2 overall, so choosing to save on computation with this observable has a clear drawback with accuracy.

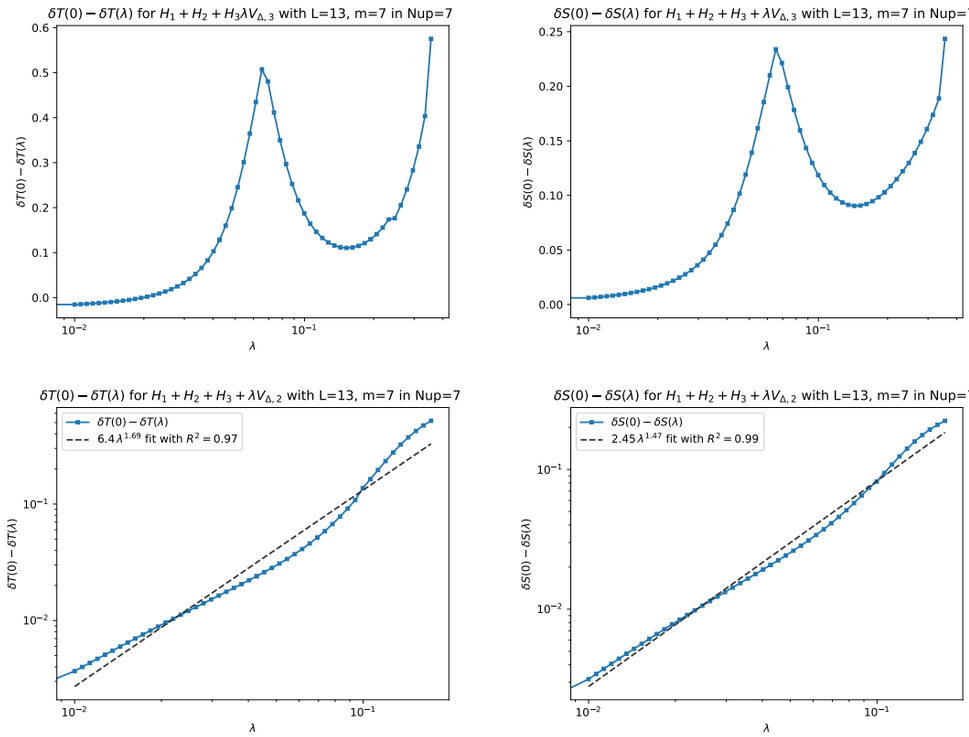


Figure 3.8: The system $H = H_1 + H_2 + H_3$ is considered for perturbations $V = V_{\Delta,2}, V_{\Delta,3}$. We see that we once again have a power law associated with $V_{\Delta,2}$, however the scar is hybridising faster than in the $H = H_1 + H_2$. As for $V_{\Delta,3}$ we observe a smooth relationship interrupted by a spike possibly due to an avoided level crossing. Despite this, it manages to hybridise slower than the previous case.

3.2.2 Perturbations

The results for $H_{\text{par}}, H_{\text{par}}$ as shown in fig.3.9 are very different, but in both scenarios it is hard to pick out a nice relationship for $\delta T(\lambda)$ and $\delta S(\lambda)$. For $V = H_{\text{par}}$ the entropy and kinetic plots behave very similarly, this is something that we observed in the previous section. The state effectively hybridises at $\lambda \approx 2$ which is a relatively huge value in comparison to the slowest anisotropy hybridisation at $\lambda \approx 5 \cdot 10^{-1}$. One could consider this to be an unexpected result, as more states resonating with the SHS scar might result in a more rapid hybridisation. Looking at $V = H_{\text{DM}}$ we observe an even slower result at $\lambda > 4$. The scar did not break within the window I calculated though it does appear to be on an upwards trend after each spike, so it's reasonable to hypothesise that it will eventually hybridise.

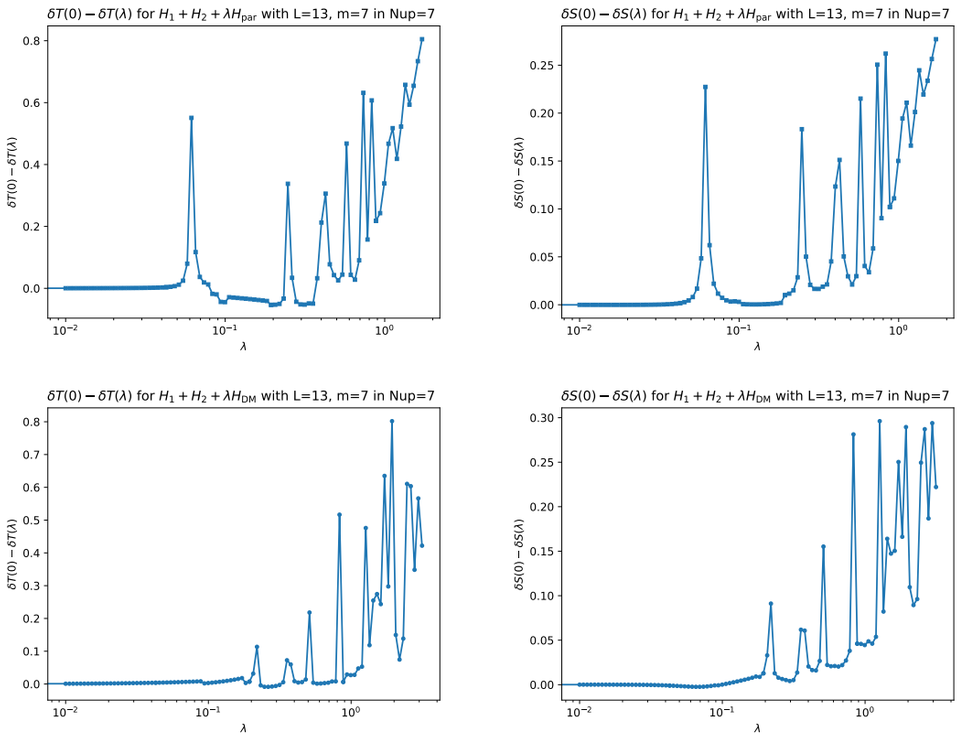


Figure 3.9: The perturbations $V = H_{\text{par}},\ H_{\text{DM}}$ are shown to eventually hybridise the scar, however at very different λ. All $\delta T(\lambda)$ and $\delta S(\lambda)$ relationships exhibit large spikes.

The dynamics around these spikes are particularly interesting. We shift our focus for a moment to the $L = 17, m = 8$ state in $N_{\text{up}} = 7$, which we've seen through our fidelity susceptibility analysis should be a particularly 'bad' state. The dynamics are as expected as can be seen in fig.3.10a, they are difficult to track, with multiple spikes. The natural question to ask is what does one of these spikes look like on the actual kinetic energy graph, the result can be seen in fig.3.10a. There seems to be another point, not associated with our original scar that has quite a low kinetic energy.

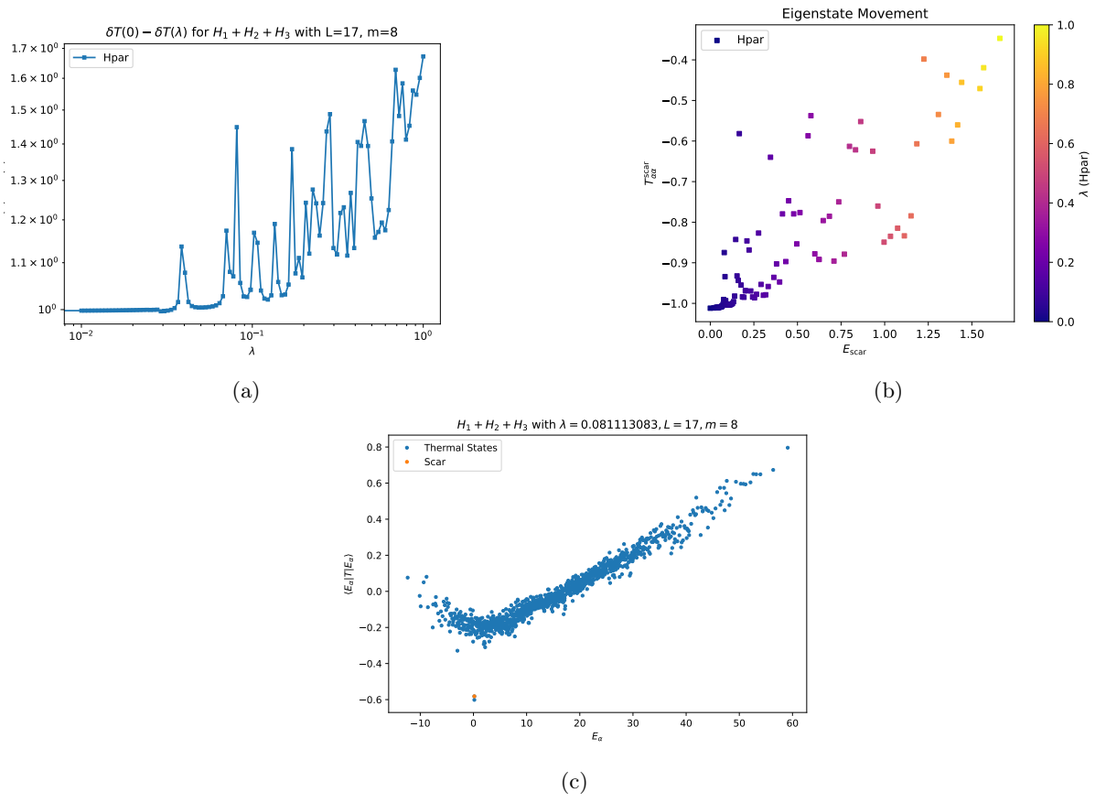


Figure 3.10: In (a) we increase λ the scar state thermalises in an interesting way, at the major peaks of the graphs it is clear that the scar is selectively hybridising with some thermal states. It's worth noting that in this plot the y axis has $\delta T(0) - \delta T(\lambda) + 1$, which was done to avoid logarithms of negative numbers. We verify our hybridising statement with (c) where it's clear that there is some nearby thermal state affecting the hybridisation of our scar. The eigenstate movement is shown in (b) to verify that we haven't travelled away from the energy dense section of the distribution.

We can investigate how exactly this close by point is related to our scar by plotting some statistics about eigenstates near in energy to our scar state. From table 3.1 we observe that the weight of the SHS has been completely split in half between two states of nearly identical energy. We can further verify that this anomalous state has not always been here by looking at results for d=0 in table 3.2. We find that indeed, we did only begin with one scar state and this overlap is purely a consequence of our perturbation.

Table 3.1: Eigenstates in the vicinity of the scar state at $d = 0.081113083$. The scar is identified by its overlap with the SHS, corresponding to eigenstate index 97.

Index	E	T_{nn}	Distance	SHS overlap
94	−0.02088471	−0.21208076	−3	5.37×10^{-4}
95	0.08940143	−0.21807468	−2	5.01×10^{-4}
96	0.14996284	−0.17243956	−1	5.44×10^{-3}
97	0.16633316	**−0.58173251**	0	$\mathbf{4.9743 \times 10^{-1}}$
98	0.16993120	**−0.60147760**	1	$\mathbf{4.9114 \times 10^{-1}}$
99	0.21108085	−0.22690450	2	2.45×10^{-3}
100	0.27336364	−0.17406969	3	1.17×10^{-4}

Table 3.2: Eigenstates in the vicinity of the scar state at $d = 0$. The scar is identified by its overlap with the SHS, corresponding to eigenstate index 95.

Index	E	T_{nn}	Distance	SHS overlap
92	−0.07606459	−0.29559810	−3	1.54×10^{-27}
93	−0.05592622	−0.16324212	−2	6.96×10^{-28}
94	−0.03180170	−0.20104421	−1	4.45×10^{-29}
95	−0.00000000	**−1.01188407**	0	**1.00000000**
96	0.07470370	−0.20844693	1	1.39×10^{-27}
97	0.13449219	−0.20091540	2	2.08×10^{-29}
98	0.15435415	−0.16885641	3	2.24×10^{-28}

Finally, it helps to provide another example of this phenomenon for a different system $L = 13, m = 6, N_{\text{up}} = 5$ so we know it likely isn't just a consequence of that particular length and magnetisation sector. In fig.3.11, we can see that the same mechanism we spotted earlier is happening again in this system. We check that the scar state indeed bounces back from the small hybridisation in fig.3.11b.

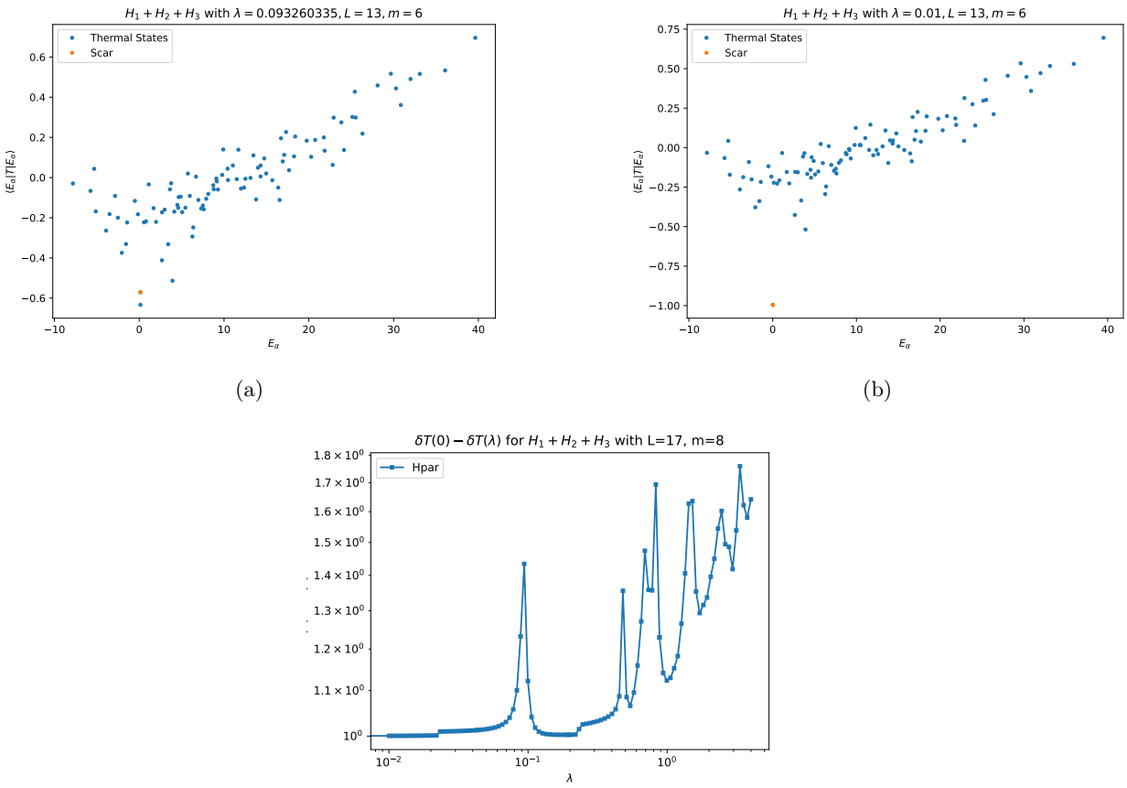


Figure 3.11: In (a) we can spot a non scar state very close to our scar, the same situation we had for our previous case. In (b) we move slightly away from this perturbation and increase it to show that unintuitively the scar comes back out as low as it was initially. In (c) we show that this shows up as a spike in $\delta T(0) - \delta T(\lambda) + 1$.

4 Conclusions

The SHS is an unusual scar state within the chaotic Heisenberg xxz model it's built from degenerate states spread across all magnetisation sectors and not confined to one symmetry block. This made it an interesting scar to study, and to perturb. The dynamics it has shown are not that of a normal ETH obeying state, but are also quite different to scar states in other systems eg. the PXP model. It has proven to be robust to certain perturbations under very specific conditions.

We find that thermal states obey the ETH well for fidelity susceptibility calculations, $\chi_F \propto e^L$. This confirms that our methodology reproduces expected behaviour. The SHS however consistently disobeys this scaling, for most L, m pairs its χ_F is below this ETH scaling, which implies this scar is somewhat stable to perturbations. However, robustness is highly dependent on L, m, N_{up}. Some choices are well behaved with low χ_F (eg. $L = 13, m = 7, N_{\text{up}} = 7$) and others are worse behaved with very high χ_F, sometimes even higher than the values predicted by the ETH. For odd L it is also the case that if χ_F blows up at some N'_{up} then it also will at $L - N'_{\text{up}}$.

Under Anisotropy, Δ_k, perturbation where $H = H_1 + H_2 + \lambda V_{\Delta,2}$, $\delta T(\lambda)$, and $\delta S(\lambda)$ fall with a close to linear power law $\lambda^{1.2\sim1.7}$. Adding higher range terms, $H = H_1 + H_2 + H_3 + \lambda V_{\Delta,2}$, makes the scar hybridise faster under the same perturbation. Generic potentials $(H_{\text{par}}, H_{\text{DM}})$ behave very differently, hybridsation does not occur til much higher λ and the $\delta T(\lambda), \delta S(\lambda)$ curves show repeated spikes rather than a smooth growth. It looks as though these spikes correspond to near resonances/avoided level crossings. The SHS weight temporarily splits with a nearby thermal state and can even 'bounce back' to almost initial $\lambda = 0$ values afterward as seen in fig.3.11, which implies that hybridsation isn't necessarily monotonic. Perturbations of the fine tuning of the anisotropy appear to destroy the scar much faster than generic potentials that share the Hamiltonian's symmetries.

The SHS behaves as a genuine, ETH-violating many-body scar under the tested perturbations, but its robustness is not a simple function of system size it depends sensitively on sector choice, and its form of hybridisation depends on the type of perturbation. An interesting route to examine in future work would be to look at the DMI (H_{DM})in greater detail and examine its directionality for weakening or strengthening the SHS.

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Appendix A: Tracking the Scar

There are multiple methods to track the scar state as it moves. The smoothest of these is an algorithm given by,

$$|\psi_{scar}(\lambda_i)\rangle = \max |\langle \psi_j(\lambda_i)|\psi_{scar}(\lambda_{i-1})\rangle|^2 \tag{A.1}$$

Where we basically choose the next state that has the biggest overlap with the previous λ step. We do this for all and should smoothly track the movement of the scar. This is the better approach to use in general for scar tracking but I was not able to use it in this paper due to the near resonant terms discussed in section 3.2. Sometimes instead of tracking the actual scar, the algorithm would smoothly attach onto the wrong state as in fig.A.1. The workaround to this issue was to use a very naive tracking method,

$$|\psi_{scar}(\lambda_i)\rangle = \max |\langle \psi_j(\lambda_i)|\psi_{scar}(0)\rangle|^2 \tag{A.2}$$

This surprisingly did smoothly track the scar, and this was verified throughout the production of the results in this report. For most states, this would not be a tracking algorithm I can recommend, however for specific situations with the SHS it proved quite useful.

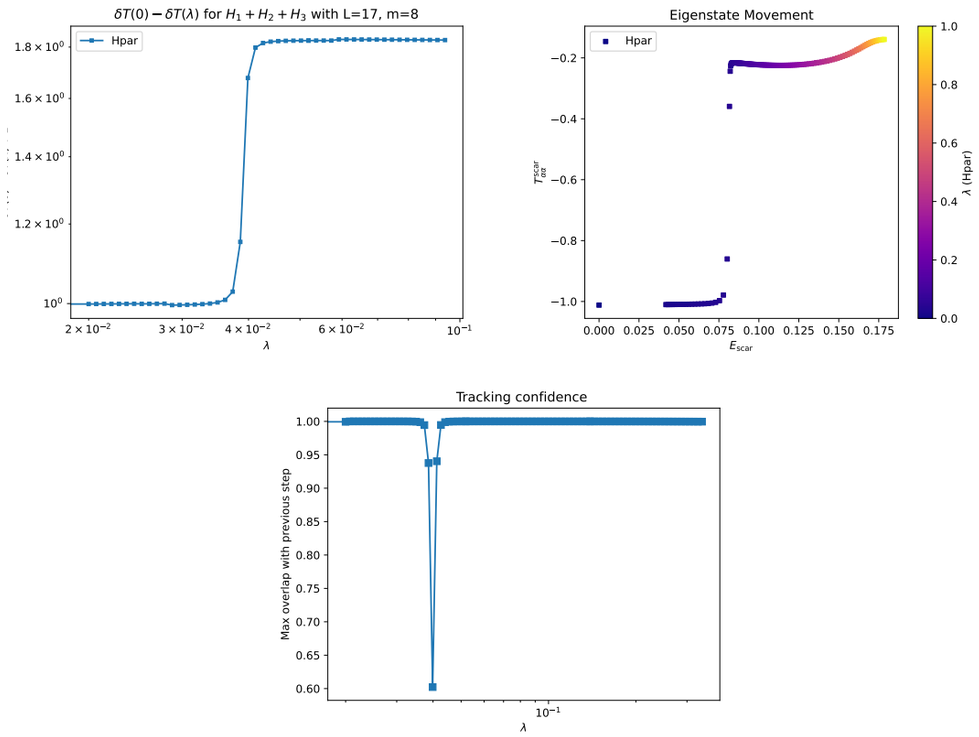


Figure A.1: In (a) in order to show the plot on a log log scale, a phase shift of 1 has been added. It's nearly as though $\delta T(0) - \delta T(\lambda)$ exhibits a phase transition at a particular value of λ. What is actually happening however is that the tracking algorithm is latching on to the wrong state, which hybridises straight away. We have shown in (b) that we indeed have tracked the scar state itself despite starting at a noticeable perturbation. In (c) it's clear that the growth has become so fast it disrupts the confidence in tracking.